

Advait Patel

advaitpatel04@gmail.com | linkedin.com/in/advait-patel | github.com/advait-patel-17 | Mobile: (309)363-5874

EDUCATION

University of Illinois - Urbana-Champaign

Champaign, IL

BS in Computer Science – GPA: 3.8/4.0

Aug. 2022 – May 2026

Relevant Coursework: Real-time Systems, Machine Learning, Deep Learning for Robotics (Grad), Algorithms, Probability/Stats, Data Structures, Systems Programming, Computer Vision (Grad), Linear Algebra

SKILLS

Python, PyTorch, C, C++, Amazon Web Services (AWS), Kotlin, OpenCV, PHP, SQL, JavaScript, Typescript, Git/GitHub, Flask, React, PyTorch, Java, Keras, Pandas, NumPy, Matplotlib, Docker, Linux/Ubuntu, ROS

EXPERIENCE

Anduril Industries

May 2025 – August 2025 (Expected)

Incoming Software Engineer Intern

Costa Mesa, CA

University of Illinois

November 2025 – Present

Undergraduate Researcher

Champaign, IL

- Exploring evaluation of robotics policies through Real2Sim using 3D Gaussian Splatting and MuJoCo.

Capital One

June 2024 - August 2024

Software Engineer Intern

McLean, VA

- Created a divisional-level dashboard to monitor problems with teams' AWS usage using Typescript, React, and AWS.
- Decreased loading time on application page by 10x by optimizing GraphQL queries.
- Implemented AWS Lambda with Python and SQLAlchemy to load data from 5 different sources into PostgreSQL.

Yield Protocol

May 2023 – August 2023

Software Engineer Intern

Remote

- Developed a Web3 chatbot using open-source large language models and Python LangChain API integrations, resulting in 20% cost reduction compared to OpenAI APIs.
- Finetuned SOTA large language models on enterprise-grade hardware with AWS Sagemaker and Huggingface APIs.
- Created a custom question-answer pipeline dataset for model finetuning, resulting in over 4,000 unique data points, which enhanced model finetuning and increased response accuracy by 35%.

Trimble

June 2021 – August 2021

Software Engineer Intern

Remote

- Developed web and mobile applications in C# to enable compliance with federal transportation regulations.

LEADERSHIP/EXTRACURRICULARS

SIGRobotics (ACM @ UIUC) | Founder, Chair | Python, ROS 2, PyTorch, MuJoCo

April 2024 – Present

- Trained pick-and-place imitation learning policies using PyTorch, LeRobot, and the Alex Koch arms.
- Developed and open-sourced LeKiwi, a low-cost mobile manipulator in collaboration with Huggingface/LeRobot.

UIUC Gies Disruption Lab | Technical Lead | Python, Huggingface, AWS

January 2023 – December 2023

- Led a team of 6 software engineers to develop an AI tutor specific to UIUC courses using finetuned open-source LLMs, vector databases, and LangChain API integrations.

Teaching Assistant

August 2023 – December 2024

- Graded and mentored student projects for CS 222 Software Design Lab and CS 361 Probability/Stats.

PROJECTS

AI Chess Pipeline | OpenCV, Python, PyTorch, OpenAI

- Developed a pipeline that takes in a top down view of a chess board and determined board state using a Resnet-18.
- Developed an AI-assisted data annotation platform using GPT-4o-mini in order to label 1.5k images.

Systems Projects in C | C

- Wrote a Unix shell from scratch, making use of fork-exec-wait paradigm.
- Implemented a text-based networking protocol in C for file-sharing, using non-blocking I/O.
- Wrote a multithreaded “make” from scratch using synchronization primitives and dependency cycle detection.